

Who's in Extreme Pain?: The Effect of Identity and Intensifiers on Physicians' Perceptions of Pain

Raef Khan

CLS 61

May 11th, 2025

Have you ever gone to the doctor and felt like you weren't being listened to?

Have you ever gone to the doctor and felt you weren't being listened to?

VIEWPOINTS

How we fail black patients in pain

Half of white medical trainees believe such myths as black people have thicker skin or less sensitive nerve endings than white people. An expert looks at how false notions and fuel inadequate treatment of minorities.

Several studies

Told After Endometriosis Surgery That Her Pain Wasn't Real, Mexico City Woman Shares Her Story to Offer Solace to Others Suffering



WILLIAM CROYLE, • Posted on Jan 7, 2025

'You are not listening to me': Black women on pain and implicit bias in medicine

Medicine has

the vestiges of which we are still grappling with today.

Racist Doctors and Organ Thieves: Why So Many Black People Distrust the Health Care System

It's more than just Tuskegee. Raci-

poisons American health care.

PUBLIC HEALTH

Black Americans expect to face racism in the doctor's office, survey finds

DECEMBER 5, 2023 · 5:00 AM ET

HEARD ON MORNING EDITION



Maria Godoy

Have you ever gone to the doctor and felt you weren't being listened to?

'We want people to take our pain seriously'

11 December 2024

Gavin Kermack

BBC News, West Midlands

Tammy Gooding

BBC Hereford & Worcester

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Background:
What's the
(linguistic) gap?



Interactions between social and pragmatic meaning is understudied

Pragmatic variables carry social information

- "He arrived at 2:58" vs. "He arrived at 3" is associated with higher intelligence but also uptightness (Beltrama 2018)



Speaker information influences pragmatic processing

- Personas impact whether “this item costs \$200” refers to the \$212 item or an unknown item (Beltrama & Schwarz, 2024)



How variable interpretations affect people in high-stakes contexts is unclear

Linguistic bias is the negative evaluation of speakers based on their voice, dialect, or linguistic variable usage

(Baugh, 2017)

This can lead to real world discrimination:



Speaking in a different dialect
→ less housing callbacks

(Baugh, 2017)



Speaking with an accent
→ less likely to be hired

(Kushins, 2014; Levon et al., 2021)



Speaking with an accent
→ less trustworthy and credible

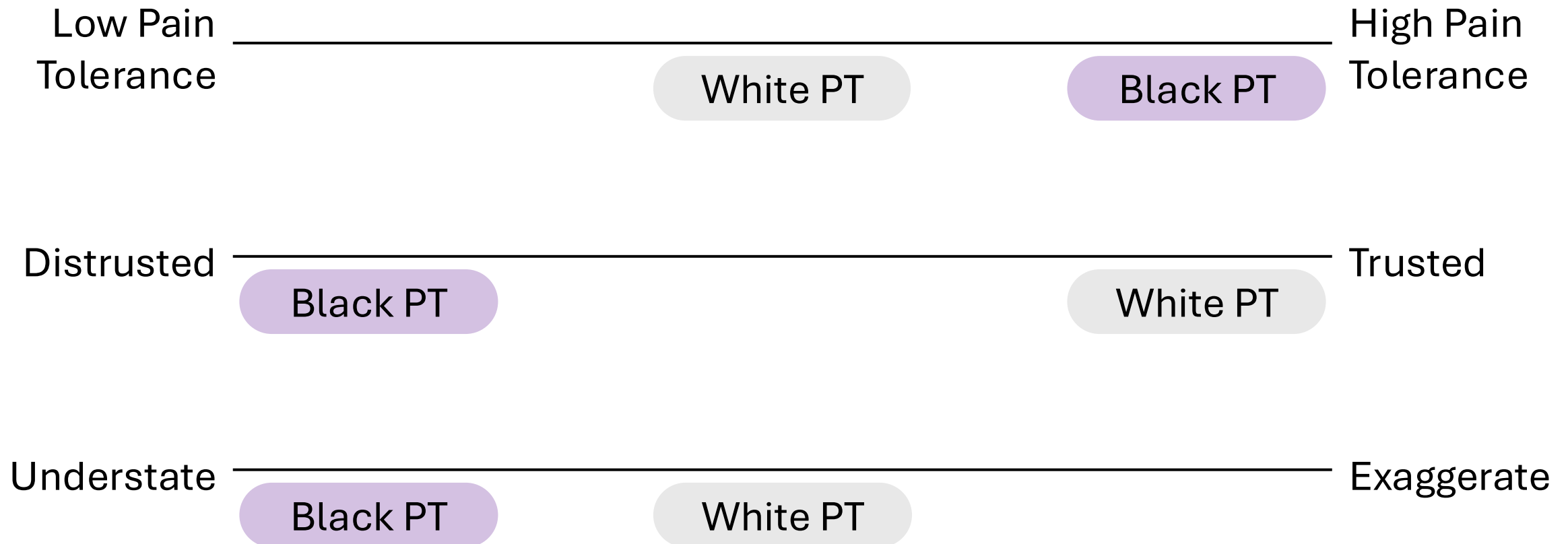
(Lev Ari & Keysar, 2010;
MacNeal et al., 2019)

Medical contexts allow for studying of socio-pragmatic variation in everyday life

Why? Studies focus on **social factors** affecting pain assessments in doctor-patient interactions

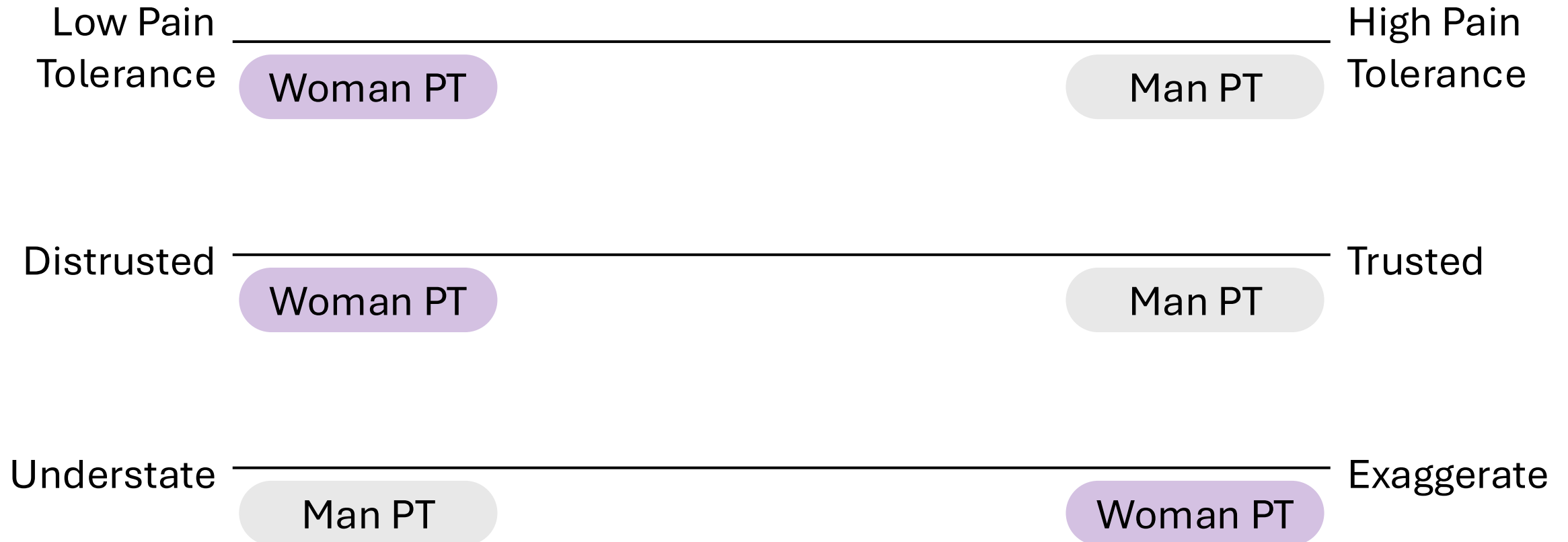
Yet few studies focus on the **linguistic content** affecting assessments (c.f. Beach et al., 2021; Kelly & Westerhoff, 2010; P. Goddu et al., 2018)

Physicians have **negative** perceptions of a patient (PT) based on the PT's race:



(Moskowitz et al., 2011; Wandner et al., 2012)

Physicians have **negative** perceptions of a patient (PT) based on the PT's gender:



(Moskowitz et al., 2011; Samulowitz et al., 2018; Wandner et al., 2012)

Medical contexts allow for studying of socio-pragmatic variation in everyday life

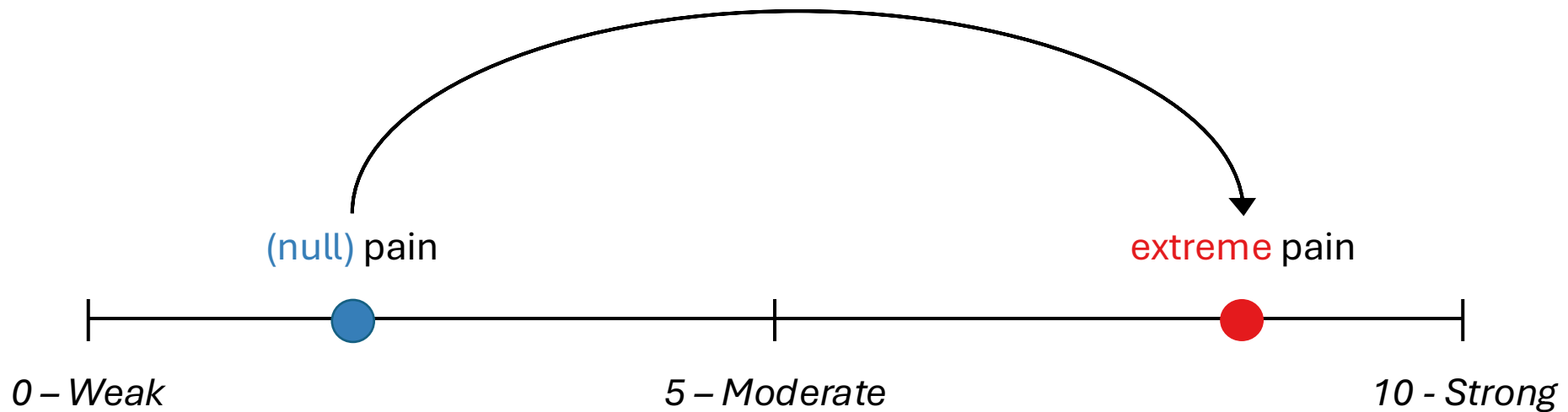
Women and Black PTs:

- are rated lower for perceived pain and receive worse care due to these negative perceptions (Guzikevits et al., 2024; Hoffman et al., 2016; Paganini et al., 2023; Ruben & Stosic, 2024)
- may ***emphasize symptoms*** in order to be heard and taken seriously (Sarwar et al., 2022)

How can PTs emphasize pain?

Intensifiers!

Intensifiers, such as '**very**' or '**extreme**', boost the meaning of a word or phrase (Ito & Tagliamonte, 2003; Kennedy & McNally, 2005)



Intensifiers also have negative perceptions, often associated with women's language and hyperbole (Burgers et al., 2016; Lakoff, 1973)

How do physicians
assess pain based on a
patient's race, gender,
and intensifier usage?

Methods: Who, What, and How?



Participants

54 MD/DO Physicians

Family/Emergency/General Medicine

US-Born and Raised

Native English speaker

61% White, 17% Black

56% Women

Recruitment Methods

Academic contacts

Listserves

Prolific



Stimuli Creation

8 conditions pulled from medical textbooks/previous study materials describing typical symptoms

Edited to read as transcript of symptom report

Verified with a medical consultant



Toshiko Uchida, MD
Director of Outpatient Simulation,
Associate Professor, Feinberg
School of Medicine

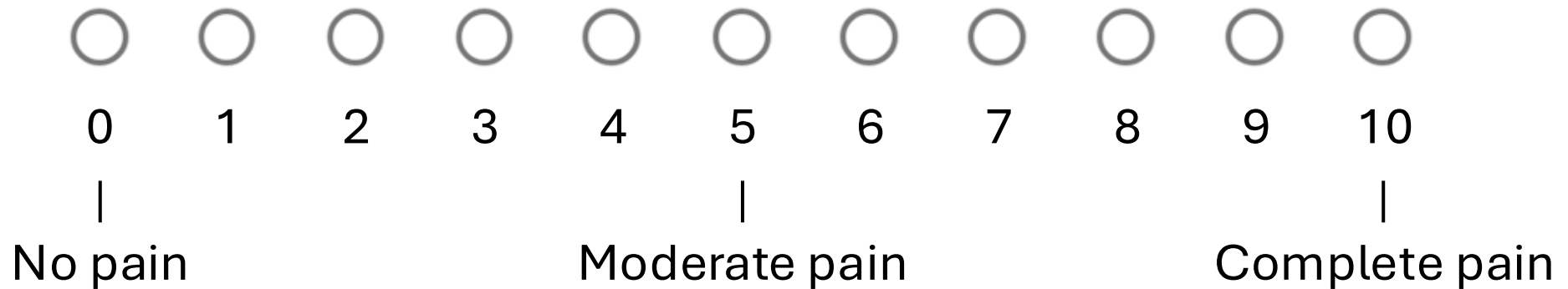
Patient Introduction

A 56-year-old Black woman comes into your office. When you ask, "What brings you in today?", the patient responds with:

I've been having extreme pain in my lower back and down both of my legs for several months. My left side hurts more, and I also sometimes have severe tingling and numbness in my thighs and legs, which makes it hard to walk comfortably. It takes intense effort to stand and lift items. It's been limiting my activity, but I have noticed it's better when I sit and bend forward or when I use heat pads. Like, for example, it's noticeably better when I lean on a shopping cart as I am walking through the grocery store. Over-the-counter medication has not been very helpful.

Trial Example: Medical vignette of a 56-year-old with Spinal Stenosis
Race: Black, Gender: woman, Intensifier: Intense

How much pain do you believe the patient is in?



Trial Example: Question about the physician's perception of the patient's pain

(Safikhani et al., 2018)

Critical Stimuli

Conditions

Spinal Stenosis: 56 y/o

Coronary Artery Disease: 60 y/o

Peripheral Neuropathy: 48 y/o

Osteoarthritis: 68 y/o

Critical Stimuli

Considerations:

Painful chronic conditions

Age restricted (45-70)

Race/gender balanced in population presentation

Manipulations:

Race: Black vs. White

Gender: man vs. woman

Intensifier usage: All **Bare** vs. **Intense**

- “Extreme”, “Intense”, “Severe”

Filler Stimuli

Considerations:

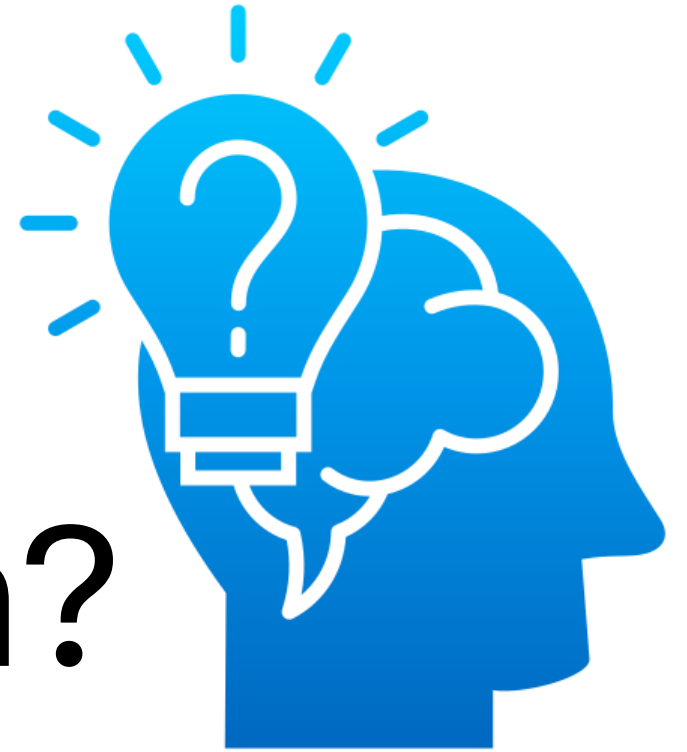
Randomized age x race x gender
!= critical trial type

Painful acute conditions

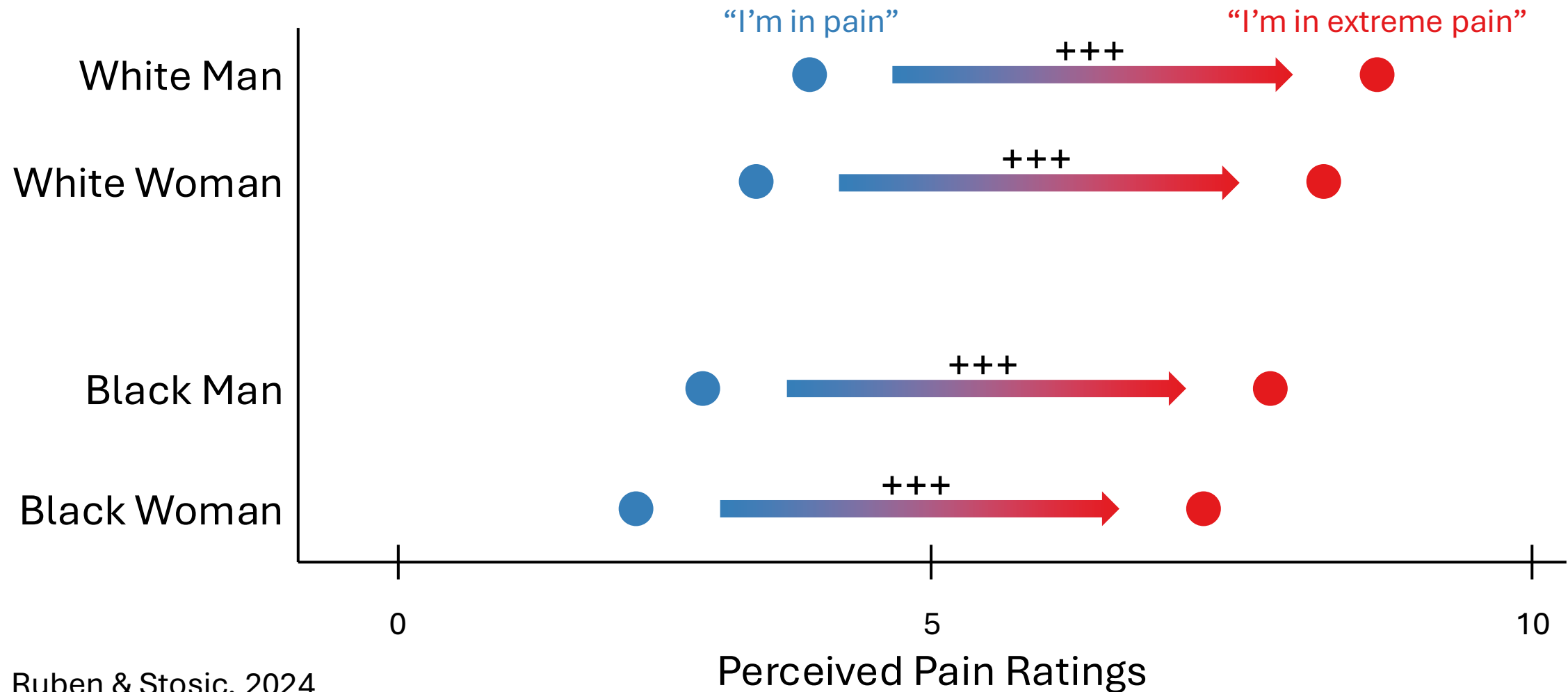
Not balanced in
population presentation

Conditions	→	Patients
Gout: 55 y/o		Asian Woman
Gastroenteritis: 35 y/o		Latina Woman
Pneumonia: 27 y/o		White Man
Appendicitis: 20 y/o		Black Nonbinary person

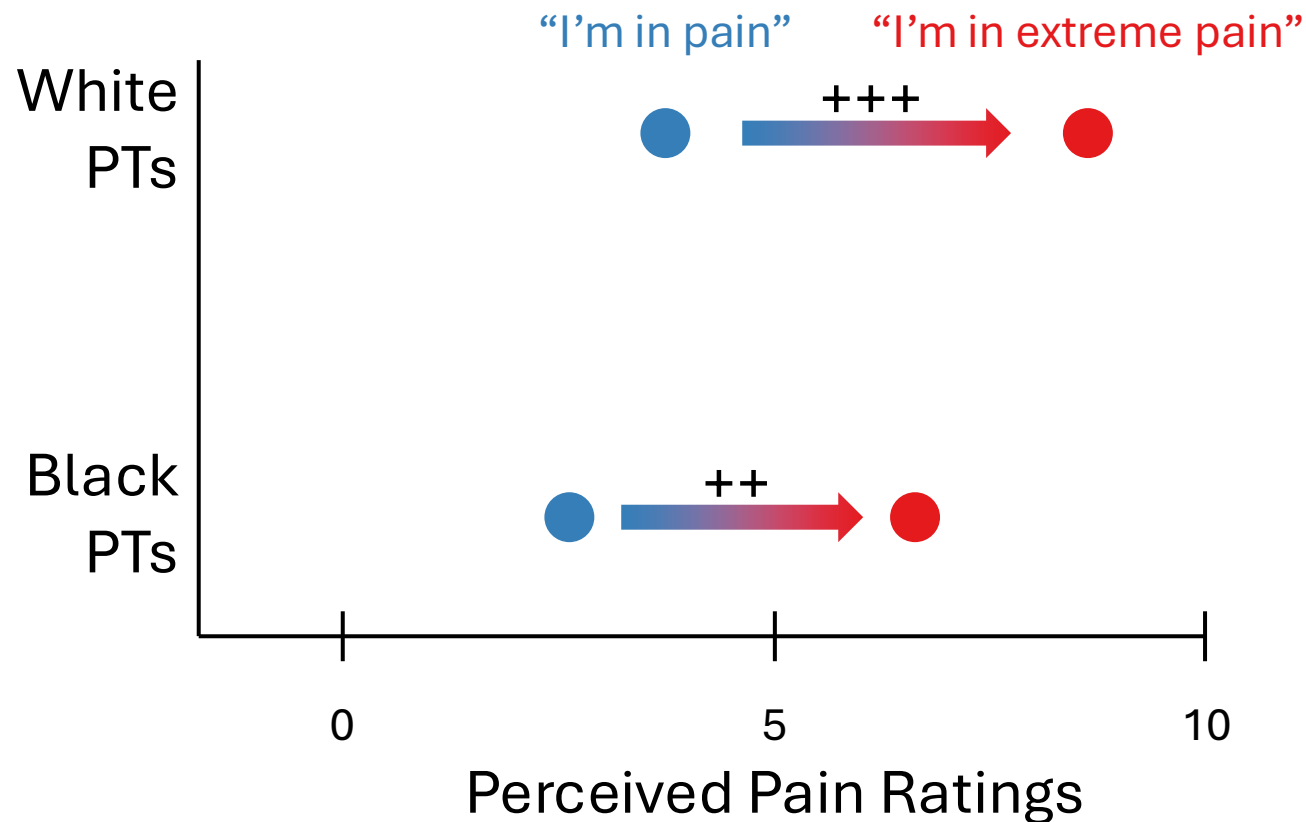
Hypotheses and
Predictions:
What might happen?



H1: Physicians' ratings of intensified pain reports is the same regardless of the PT



H2: Physicians' ratings of intensified pain reports is influenced by the PT's race

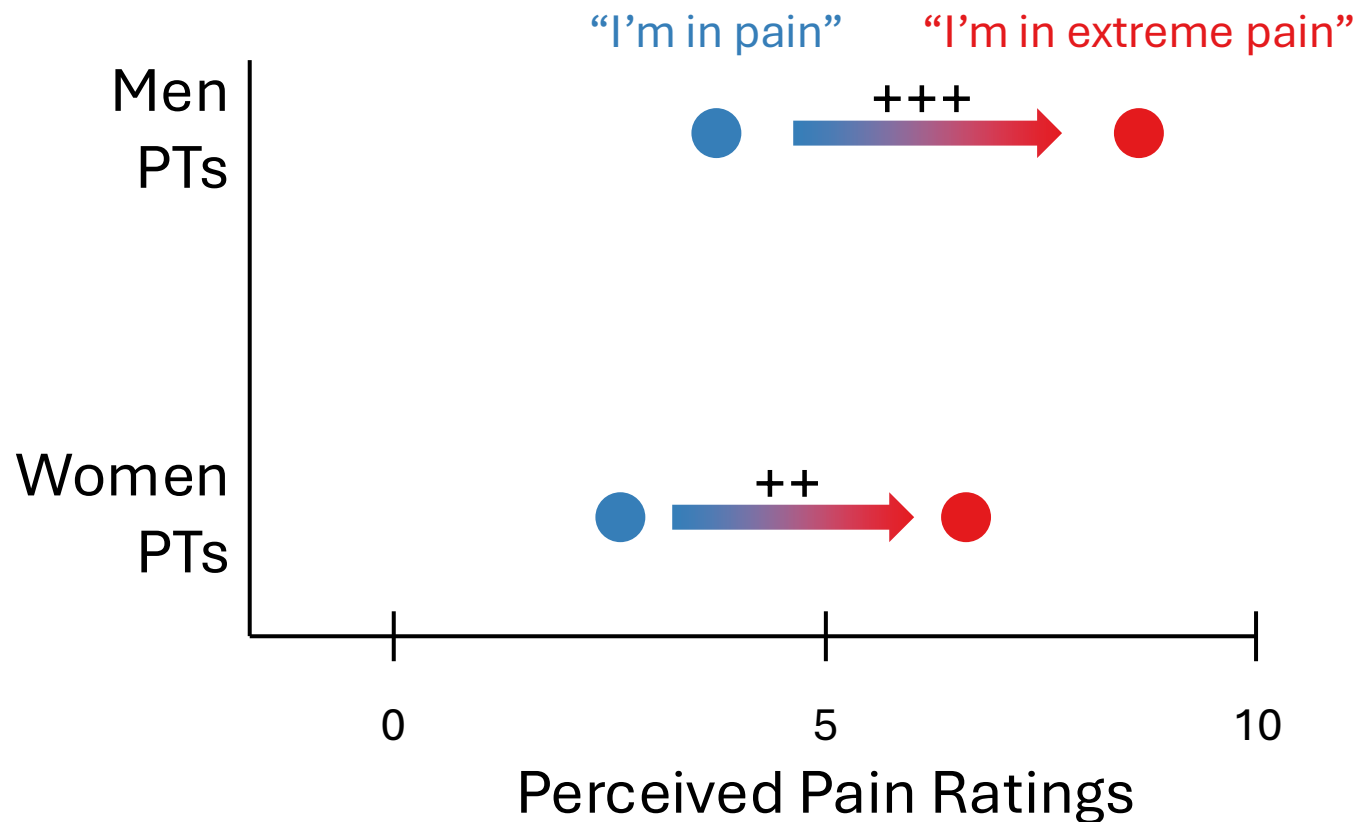


Black PTs are not trusted with their pain reports...

(Moskowitz et al., 2011)

Intensifiers may trigger a smaller boost.

H2: Physicians' ratings of intensified pain reports is influenced by the PT's gender

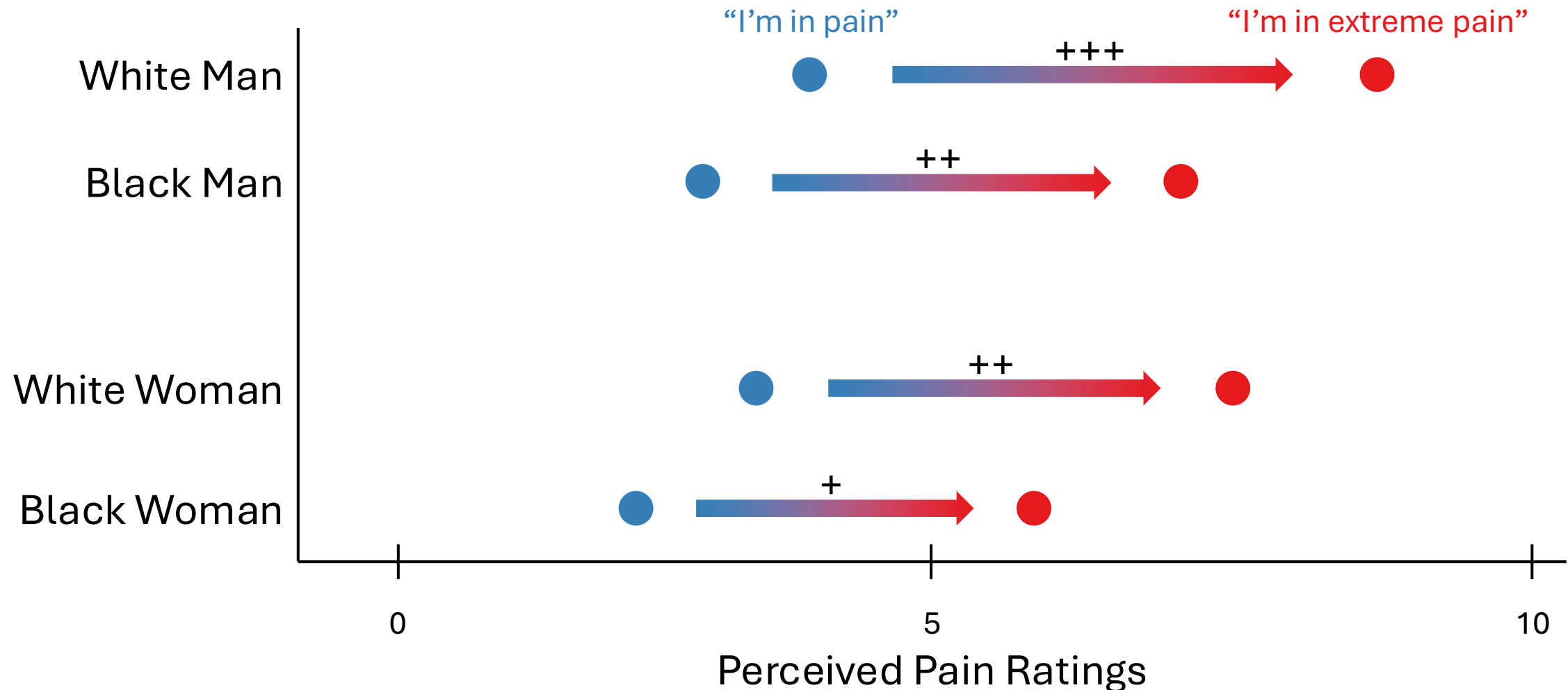


Women PTs are thought to exaggerate their pain reports...

(Samulowitz et al., 2018)

Intensifiers may trigger a smaller boost.

H2: Physicians' ratings of intensified pain reports is influenced by the PT's identity



Results and Discussion: What did we find?



Statistical analyses used ordinal regressions

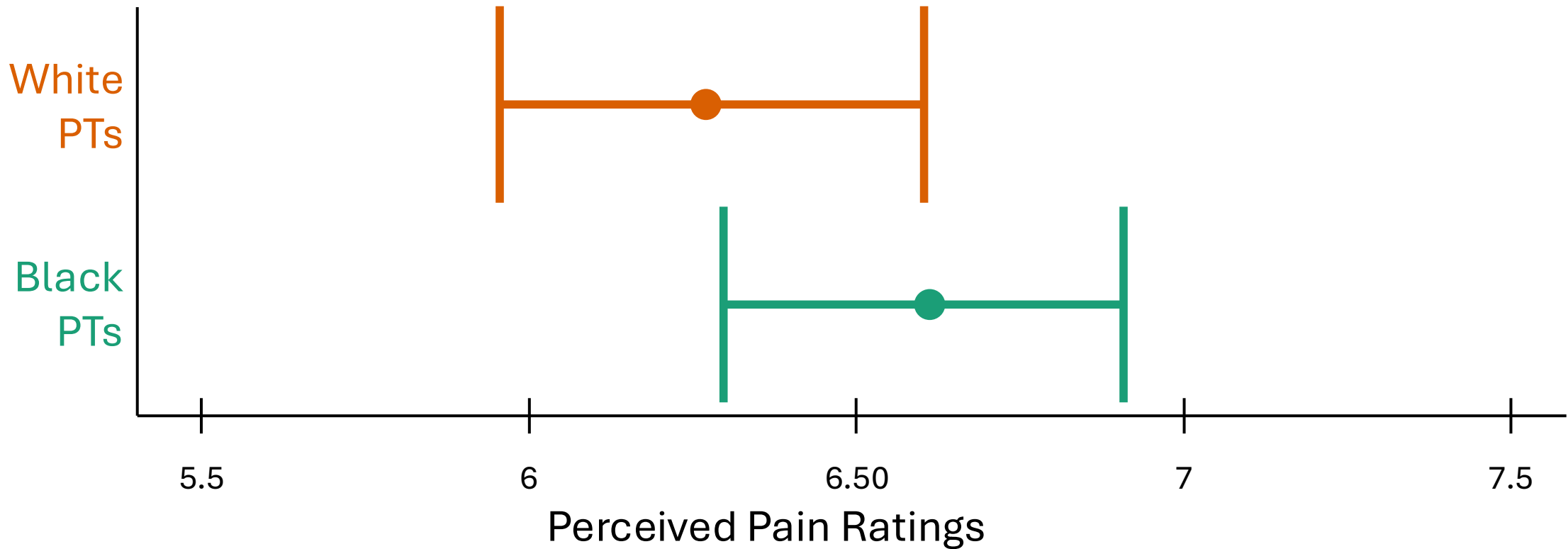
Ratings ~ Race * Gender * Intensity + (1|Participant) + (1|Item)

- White, Man, Bare coded as = +.5
- Black, Woman, Intense coded as = -.5

Likelihood Ratio Test used to determine predictor significance in model

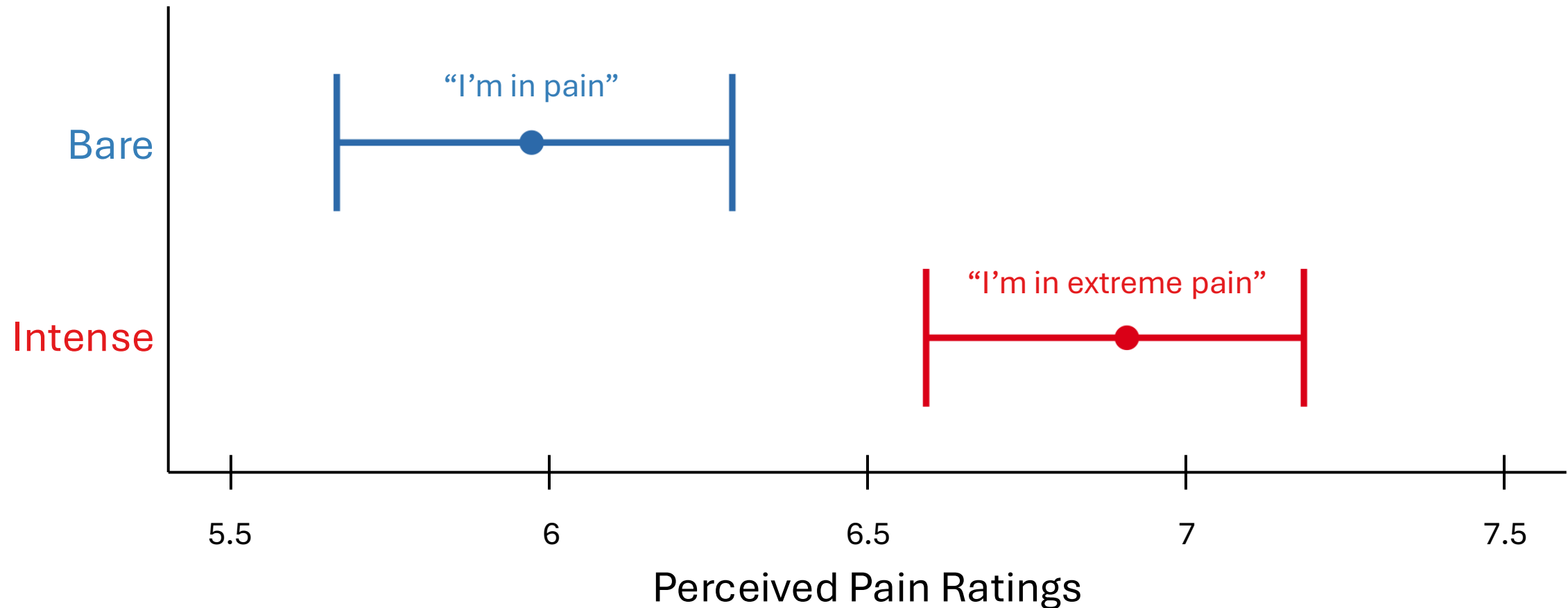
Posthoc Tukey HSD pairwise comparisons conducted for interactions using Emmeans

Physicians rate **Black PTs** pain *higher* than **White PTs** ($p < .05$)

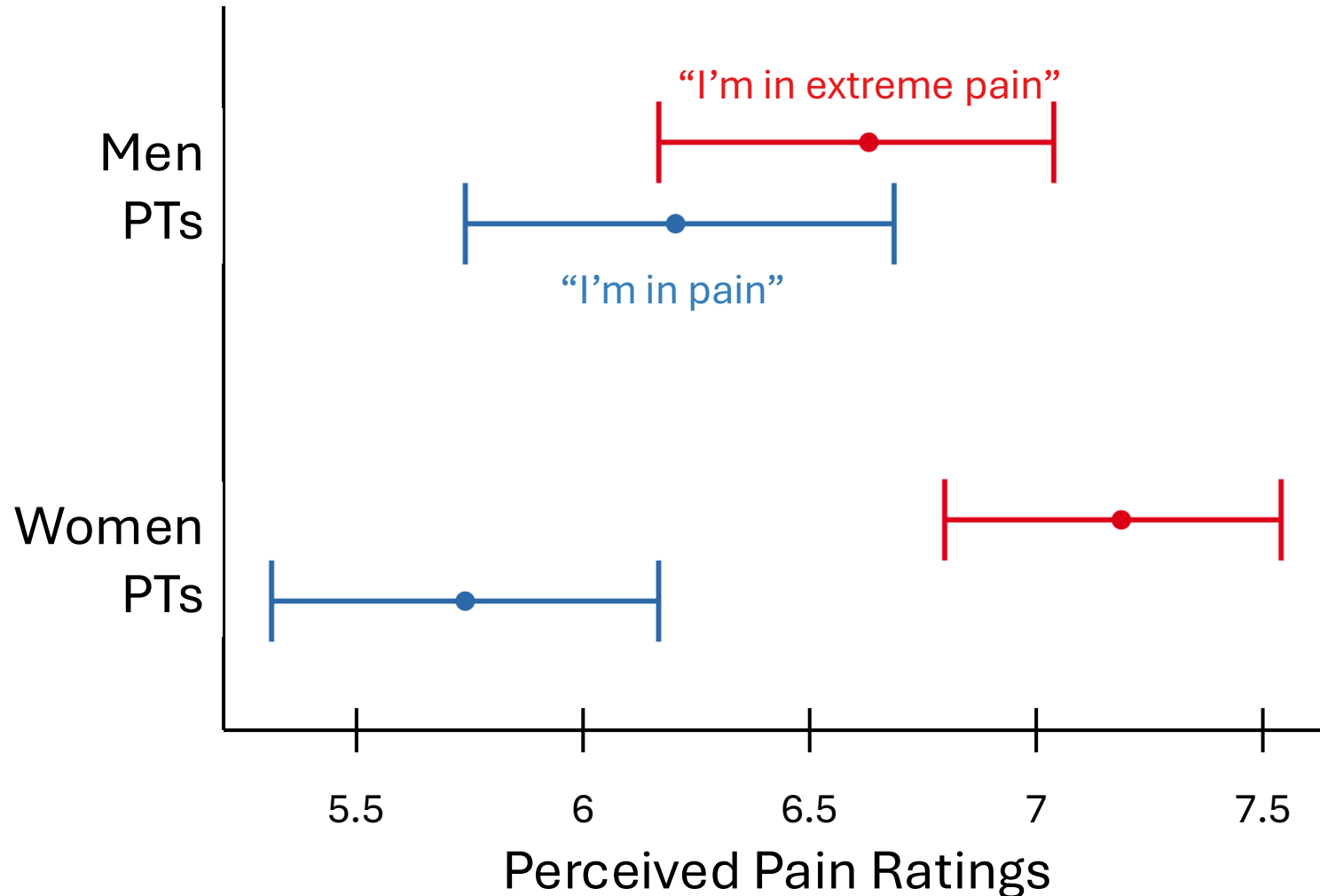


Physicians may be aware of racial bias and are overcompensating (Hoffman et al., 2016)

Physicians rate **Intense** pain reports *higher* than **Bare** pain reports ($p < .05$)

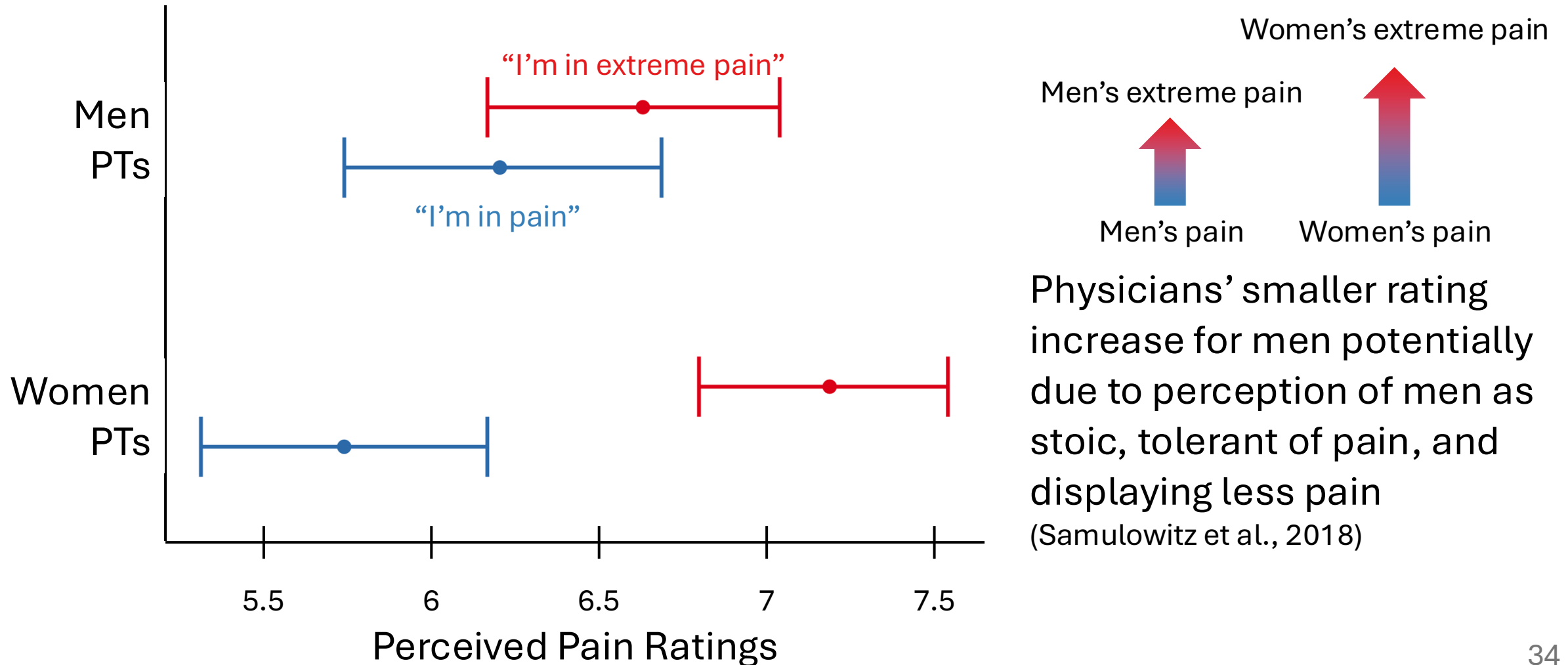


Physicians rate pain differently based on intensifier usage interacting with gender

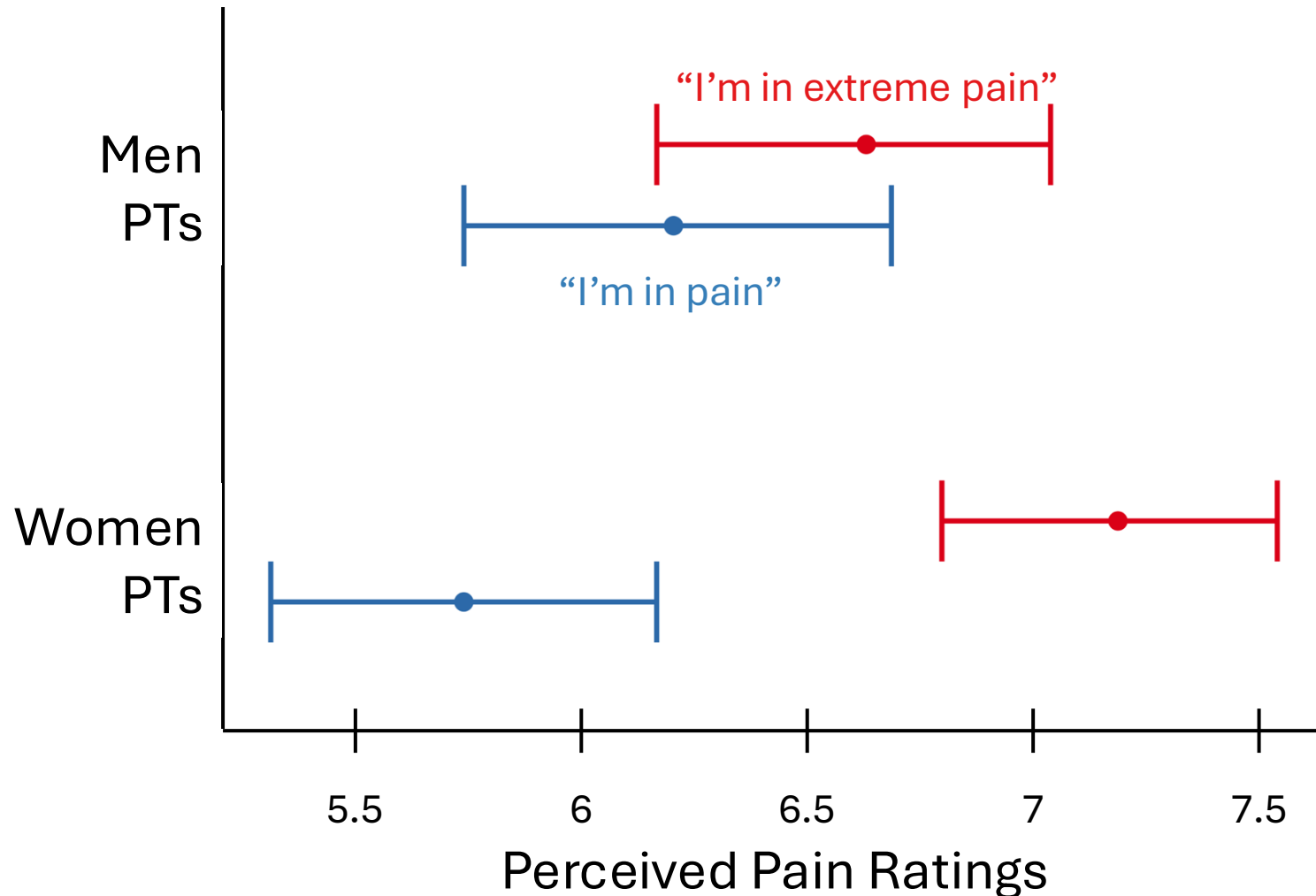


Intense vs. **Bare**
rated ***significantly different for women*** but not men ($p < .01$)

Various social biases may have influenced intensifier interpretation



Various social biases may have influenced intensifier interpretation

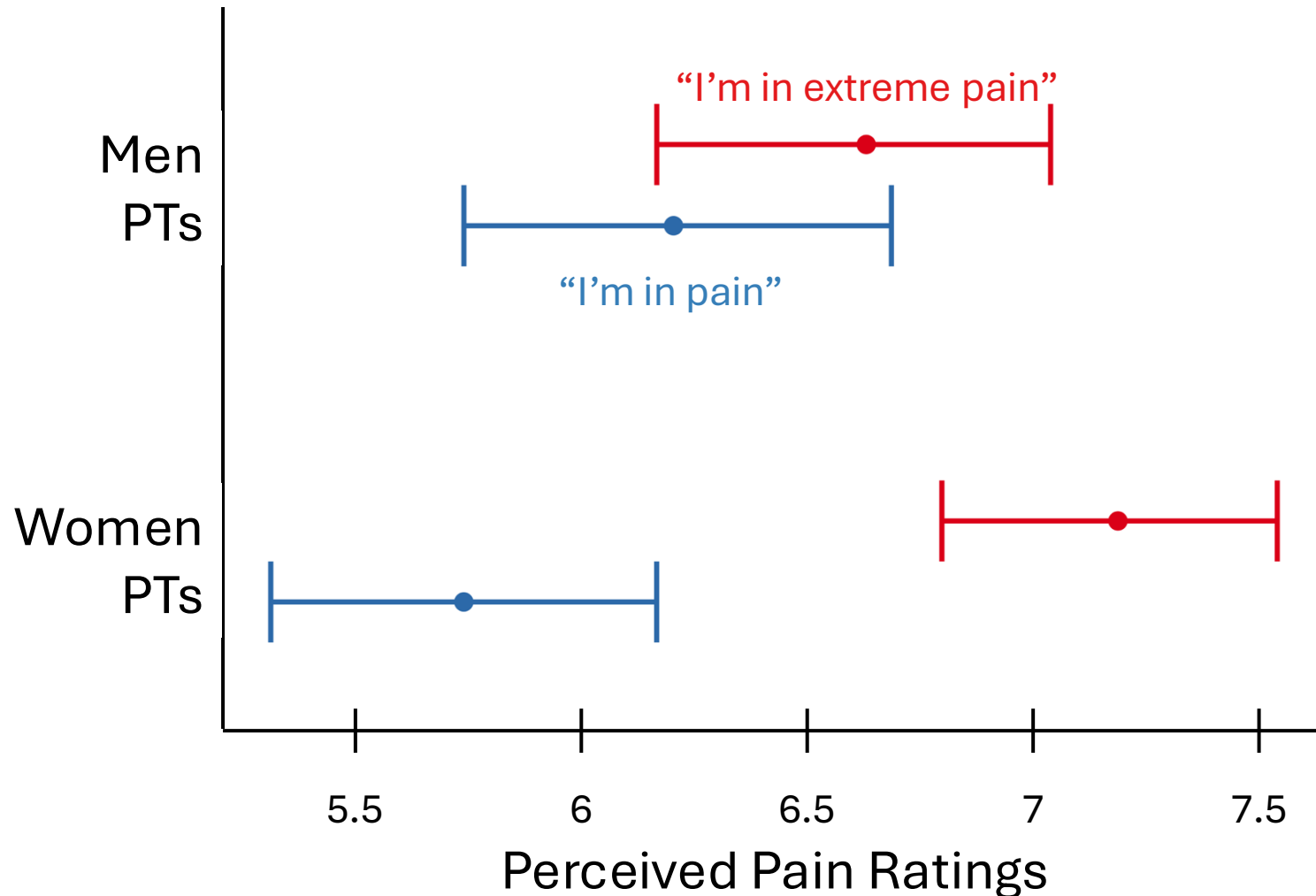


Women:
"I'm in pain"

5.74/10

Physicians potentially rated women in even less pain without intensifier usage due to expectation of intensifiers from women linked to pain reports (Samulowitz et al., 2018; Tagliamonte, 2008)

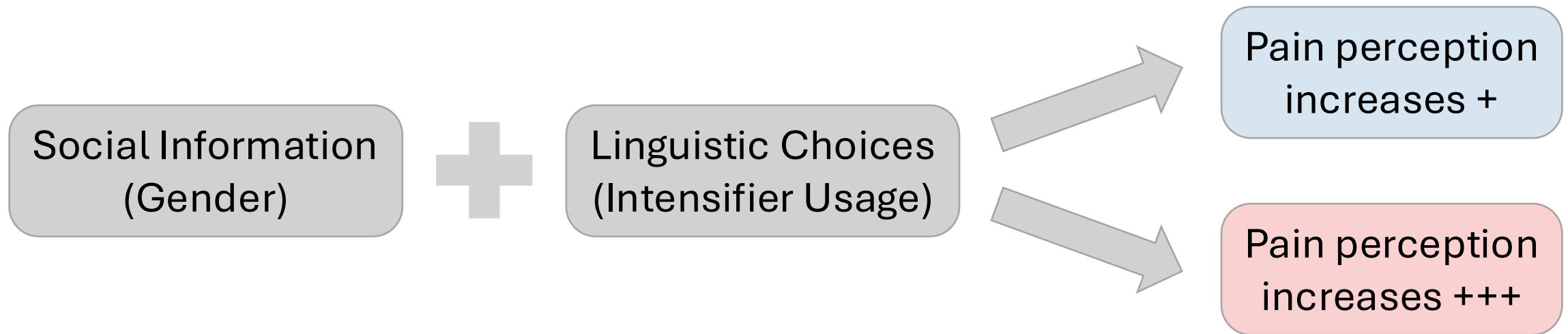
Various social biases may have influenced intensifier interpretation



Women:
"I'm in extreme pain" **7.19/10**

Physicians generally disbelieve and dismiss women for their pain (Paganini et al., 2023), potentially took women more seriously only when they emphasized pain.

Social information *does* impact pragmatic interpretations



Social factors should be incorporated into theories of linguistic meaning

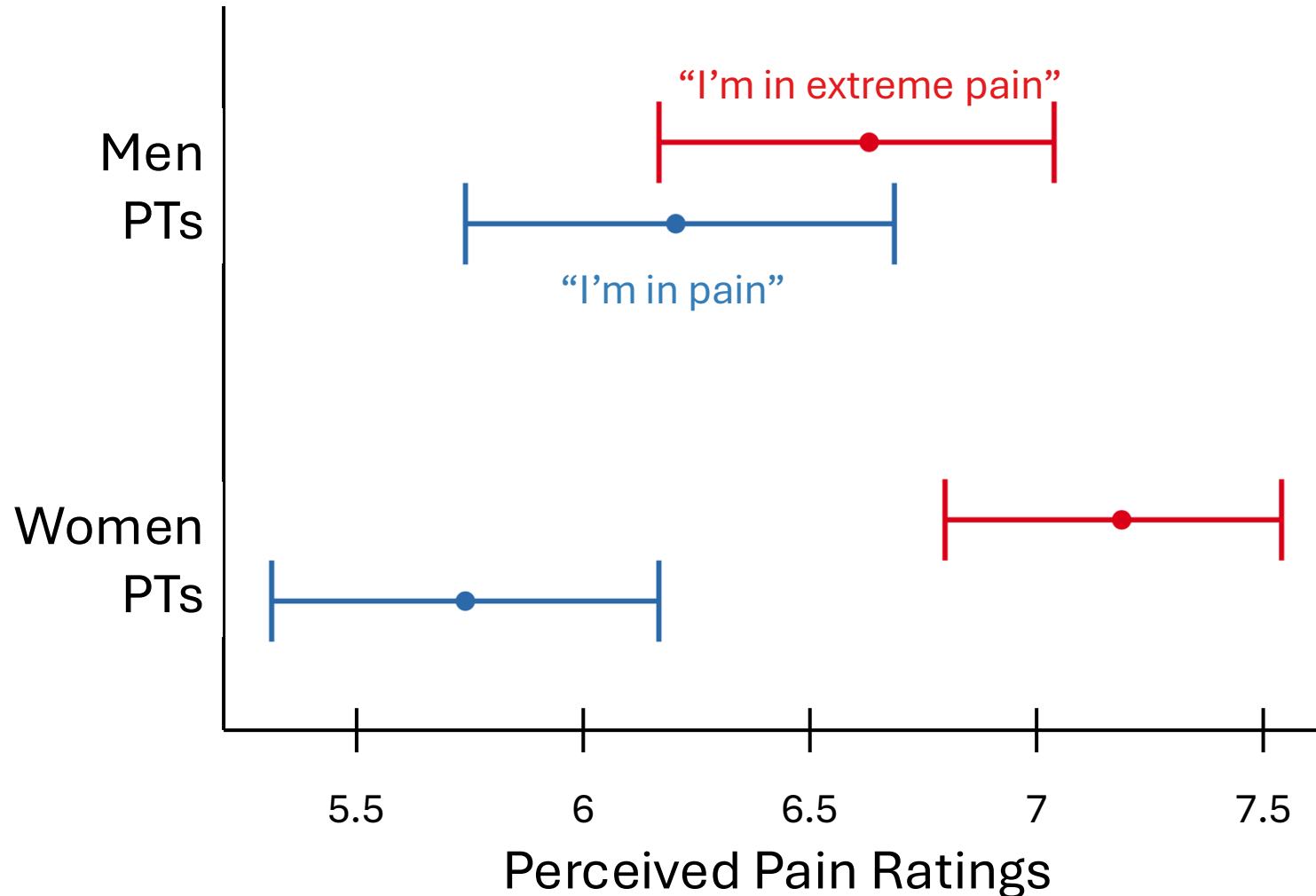
Thank you!

Thanks to the following people/groups for help with this project:

- Annette D'Onofrio
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- Toshi Uchida
- Sociogroup
- ExMeanGroup
- NWAV 51

... and others!

Questions? (or email me!)



Intense vs. Bare
rated ***significantly different for women*** but not men ($p < .01$)